



NDNts video streaming using QUIC and WebTransport

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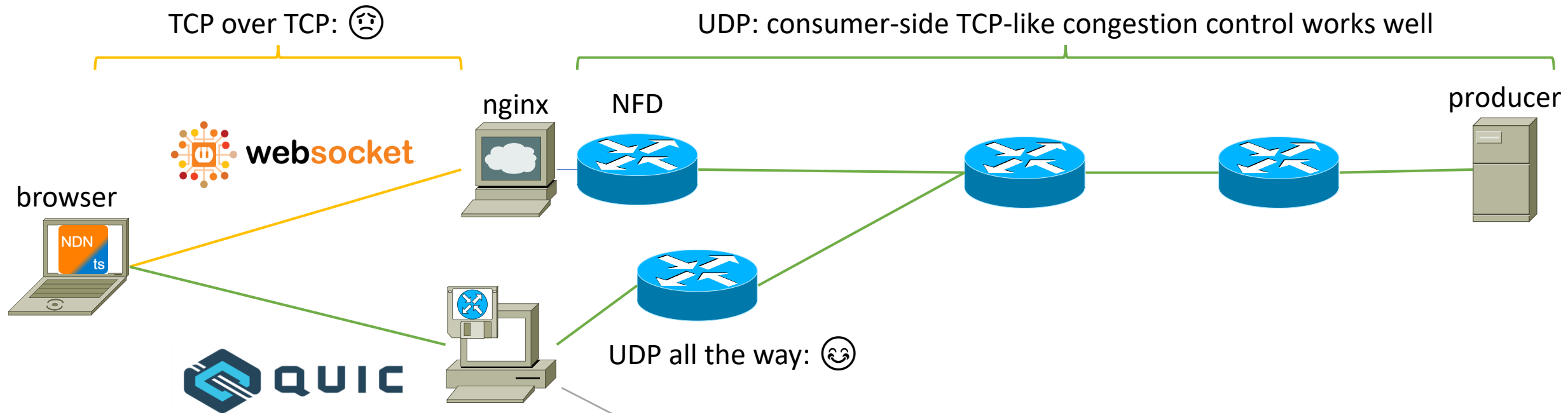
NDN Community Meeting 2021

NDNts-video – a brief history

1. Developed in parallel with iViSA to test NDNts congestion control.
 - Same streaming library: Shaka Player & Shaka Packager.
2. Publishing my own content since Black Friday 2020.
 - Content attracts viewers, not technology.
 - DASH format; four resolutions: 240p, 360p, 480p, 720p.
3. Experimented with QuicTransport in Chrome browser.
 - Data collection period: 2021-02-12 ~ 2021-03-13



WebSocket → QUIC



QUIC datagrams (draft-ietf-quic-datagram)

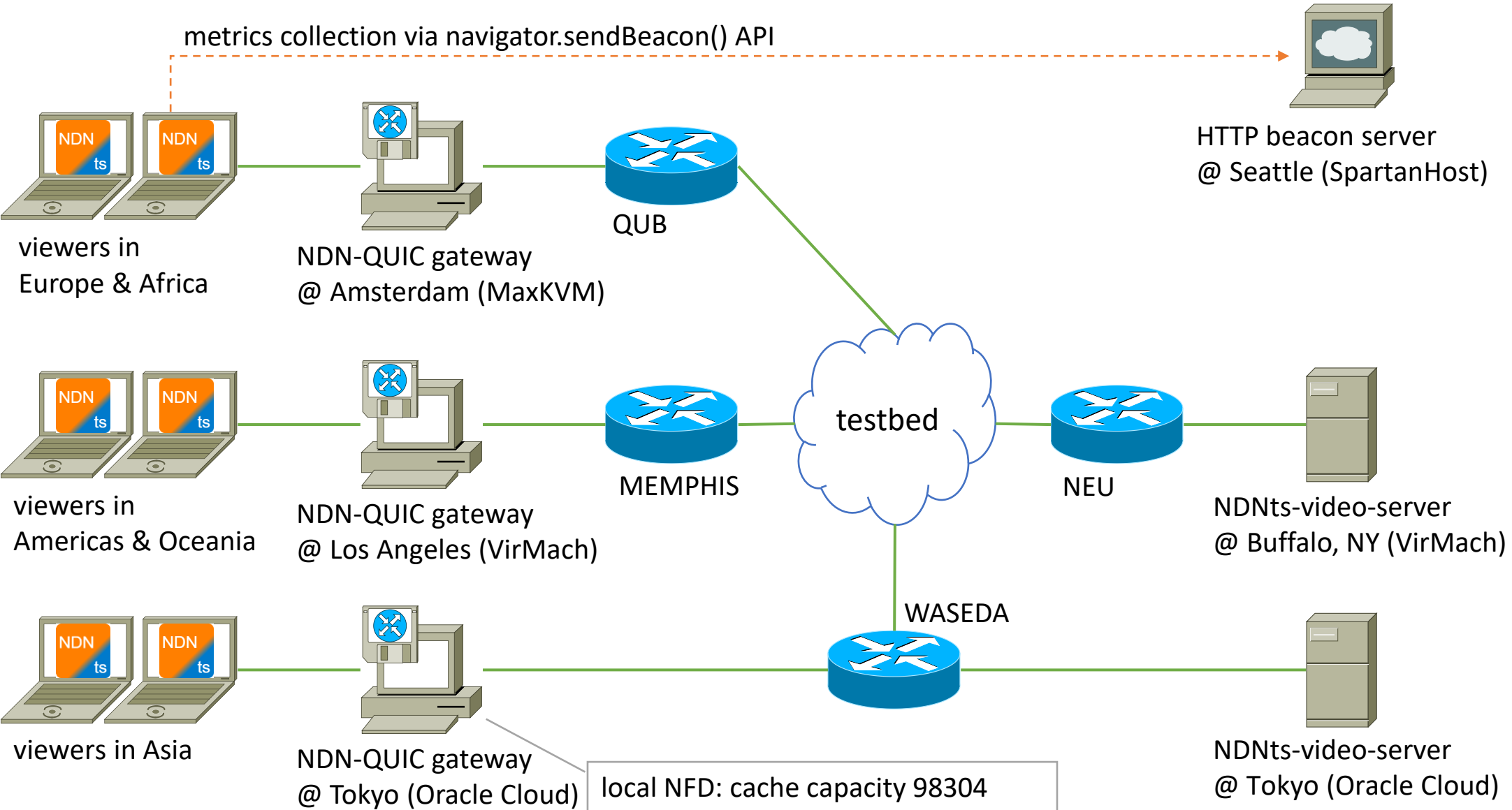
- unreliable datagrams (just like UDP)
- encrypted and authenticated

Typical MTU: 1200

NDN-QUIC gateway

- built with aioquic & python-ndn
- proxy between QUIC and plain UDP
- each QUIC connection is a UDP face on local NFD
- local NFD provides forwarding and caching

NDN-QUIC Deployment



"NDN push-ups", real world viewers

<https://pushups.ndn.today>

1. Posted on LowEndTalk & LowEndSpirit forums, became a meme.
 2. Members click and watch my push-ups.
 3. I get real world usage statistics.
 - 541 viewers from 6 continents, in 32 days
 - 6.4 hours total playback time
- Uncontrolled, but more realistic than iViSA:
 - iViSA runs emulated viewers in AWS.
 - Most real viewers don't have data center quality network connection.



yoursunny Member, IPv6 Advocate
December 2020



Push-ups solve all the problems. Serve push-up videos from all your VPS. Never leave a VPS idle again!

"sophisticated troll"



Lyphiard Member, Provider
April 15

@SirFoxy said:
what if i do 15 pushups

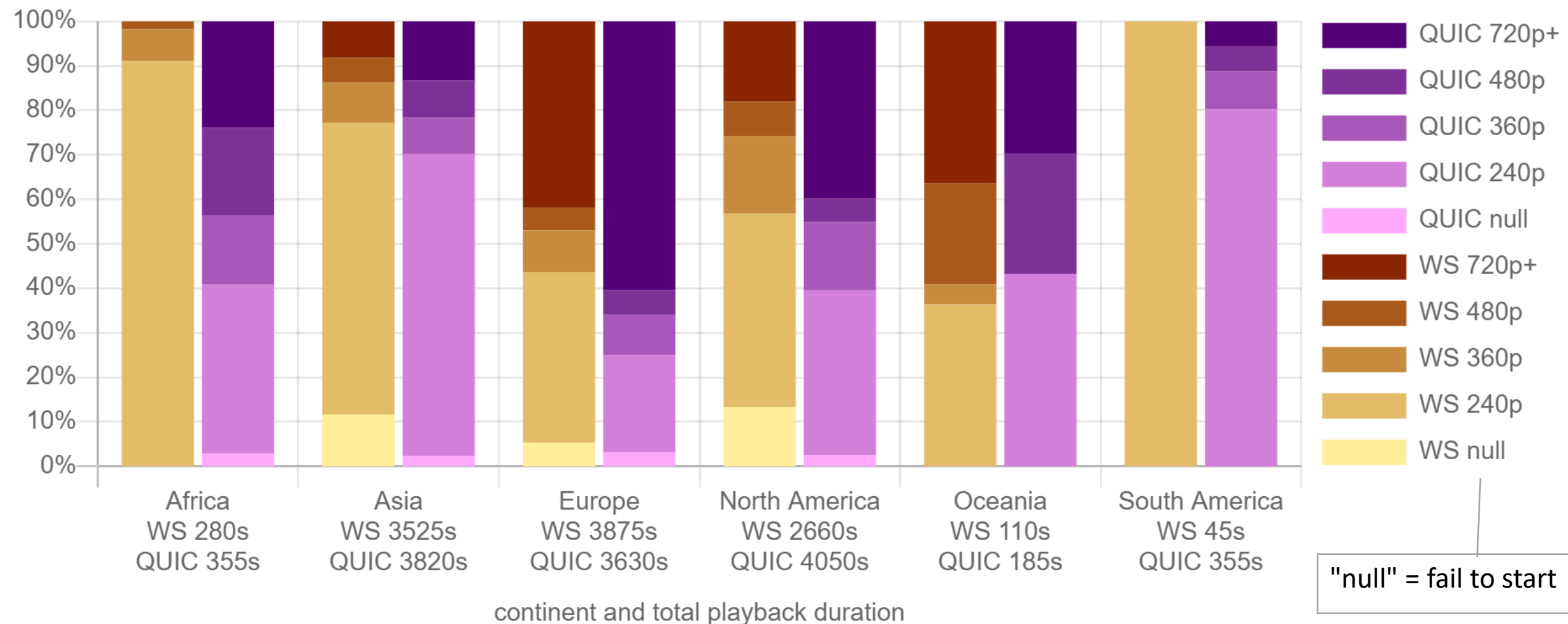
You can have \$15 off the normal yearly price of \$32.40

Video Resolution

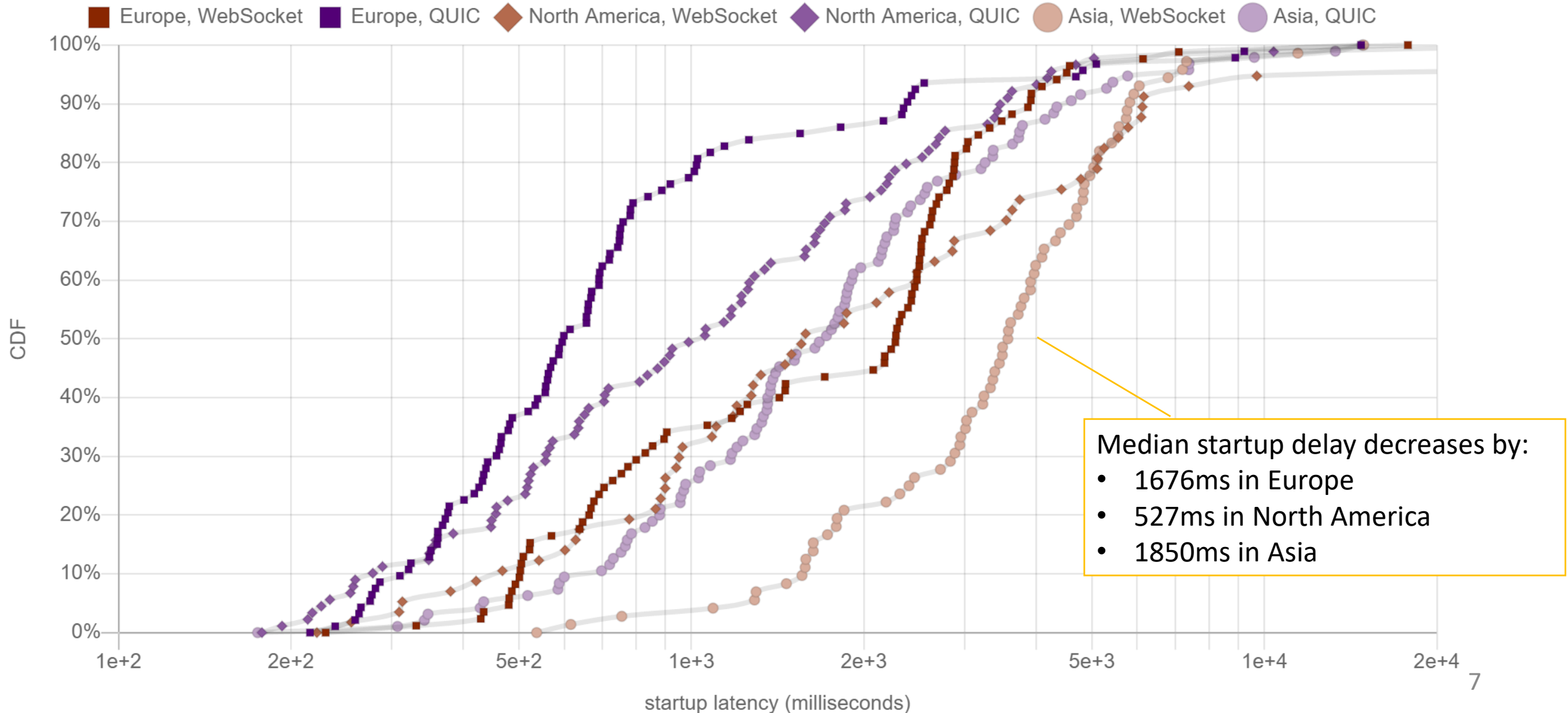
Europe, time playing at 720p resolution:

- WebSocket: 42%
- QUIC transport: 60%

North America and Asia have similar improvements.

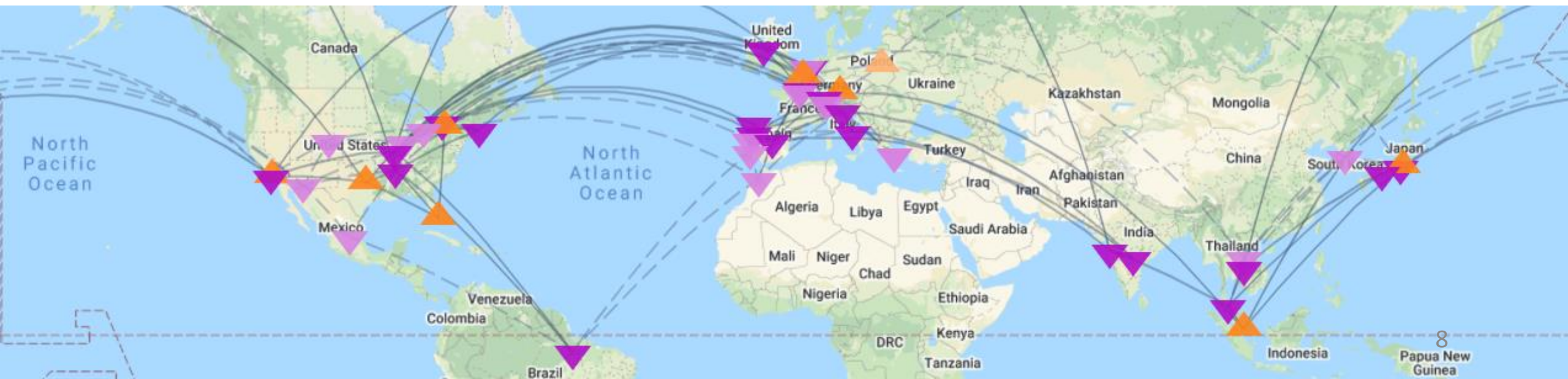


Startup Delay (🎬 → 📺)



What's Next for NDNts-video?

- QuicTransport replaced by WebTransport, using HTTP/3 over QUIC.
 - Origin Trial in Chrome 91-94 (ended).
 - Shipping in Chrome 97 (2022 Jan). Shall we support it on testbed?
- I'm building ndn6, my own global scale NDN network.
 - 9 routers. Peering with NDN testbed. IPv6 enabled. WebTransport supported.
 - Video producers moved there, still improving.



Q&A

- Download the slides: <https://talks.ndn.today>
- Watch the push-ups: <https://pushups.ndn.today>
- Read the source code:
 - <https://github.com/yoursunny/NDNts-video>
 - <https://github.com/yoursunny/NDNts-video-server>
- Extended analysis on my blog:
 - <https://yoursunny.com/t/2021/NDN-video-reality/>
 - <https://yoursunny.com/t/2021/NDN-video-QUIC/>
 - <https://yoursunny.com/t/2021/NDN-video-ndn6/>
- WebTransport introduction: <https://web.dev/webtransport/>